Invention Disclosure Form

Title

Adaptive Stair-Climbing Vacuum Cleaner

Brief Abstract

A stair-climbing vacuum cleaner that autonomously ascends/descends staircases while performing vacuuming and mopping operations. The system uses two independently actuated drive units on either side of the stairs connected by an adjustable frame; computer vision (camera/LiDAR) scans each step and an AI control unit adapts frame length and motion profile for varying step heights and depths. A movable mop/vacuum module positioned between the drive units cleans each tread as the unit progresses. The device combines powerful suction, edge brushes, omnidirectional/omni-wheel mobility for horizontal repositioning, and a retractable rod/frame mechanism for stair negotiation, enabling automated, stable, energy-efficient stair cleaning and obstacle avoidance.

Key Words

Stair-climber, Robotic vacuum, Adaptive frame, Computer Vision, LiDAR, AI navigation, Machine Learning, Microcontrollers, Sensors , Omnidirectional wheels.

Prior Art

Known devices:

- Robotic vacuums (iRobot Roomba, Roborock, Ecovacs): avoid stairs, cannot clean stairs.
- Stair climbers / mobility devices (Liftkar PTR, Scalamobil): can climb stairs but are bulky, low speed, for human transport not cleaning.
- Advanced robots (Boston Dynamics Spot, Honda E2-DR): prototype/expensive, not targeted to household stair cleaning.

Deficiencies: Inability to combine autonomous stair negotiation and effective cleaning (vacuum + mop), poor adaptation to variable step geometry, weak suction in mobile stair contexts, high cost/complexity. None perform simultaneous stair climbing and cleaning. Lack of adaptability for varied stair dimensions and mixed surface types. Excessive size, high manufacturing cost, and complex maintenance. Limited battery efficiency and absence of integrated wet-and-dry cleaning.

How this invention overcomes them:

- Dual drive units with an adjustable connecting frame allow mechanical adaptation to multiple step sizes while maintaining stability.
- Computer vision + LiDAR for real-time step geometry detection enables frame adjustment and optimized gait for each step.
- Centralized cleaning module (movable mop + strong suction) placed between drive units focuses cleaning power on each tread.
- Use of omni-wheels and intelligent path planning for efficient horizontal repositioning and obstacle avoidance.
- Design targets affordability by using modular, mass-manufacturable components and energy-efficient motors.
- Performs climbing and cleaning concurrently, reducing cleaning time.
- Compact, modular structure suitable for both stairs and flat floors.
- Dual-mode cleaning (vacuum for dust, mop for liquid) operating simultaneously.
- Adaptive frame and AI-based control enable scalability for multi-floor facilities.
- Lower energy consumption through optimized motor and power regulation.

Problems Addressed & Limitations

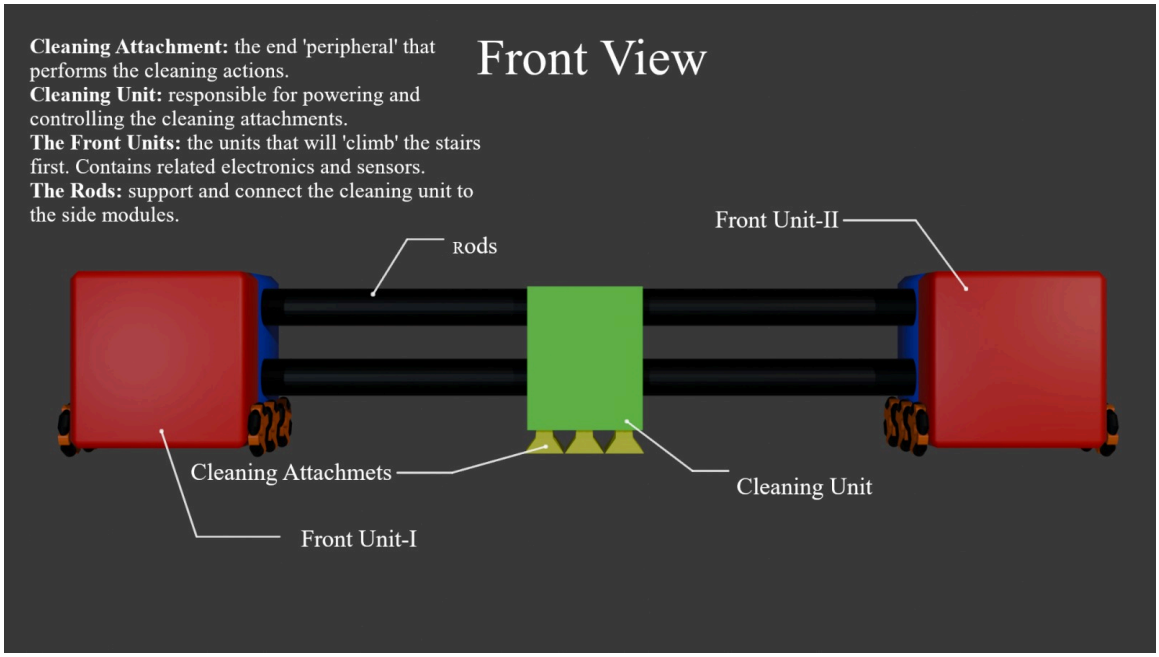
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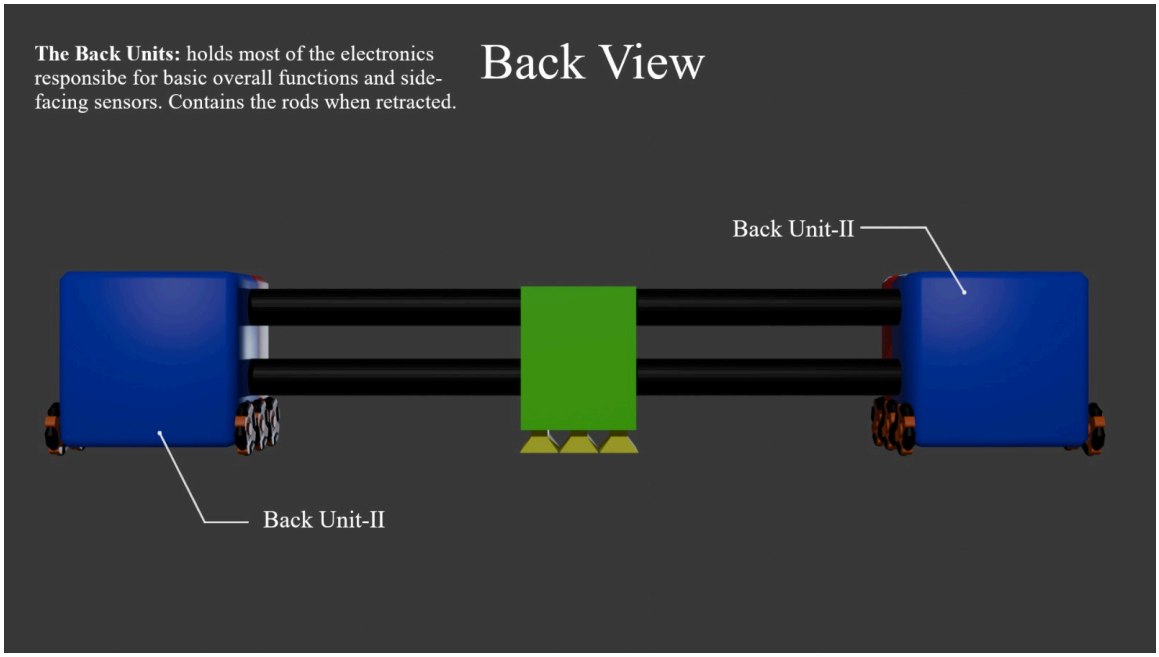
1. Enables autonomous cleaning of stairs (no manual intervention).
2. Adapts to varying step heights/depths/materials via vision/LiDAR.
3. Provides both vacuuming and wet mopping for complete step cleaning.
4. Balances suction power and mobility for effective debris removal.
5. Designed for cost-effective manufacture compared to high cost mobility robots.

Limitations (known/anticipated):

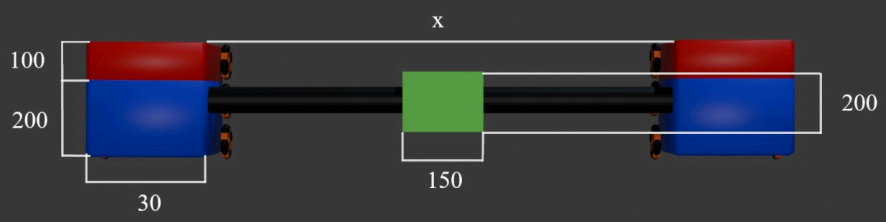
- Extremely irregular, damaged, or nonstandard stairs (very steep, broken treads) may still require manual assistance or fallback behaviors.
- Battery life will be lower during continuous stair climbs with high suction so docking/charging strategy or replaceable battery packs are required.
- Very narrow stairs where the device width cannot fit will not be supported.
- Precise performance depends on sensor calibration and control tuning for specific stair geometries.

Detailed Description

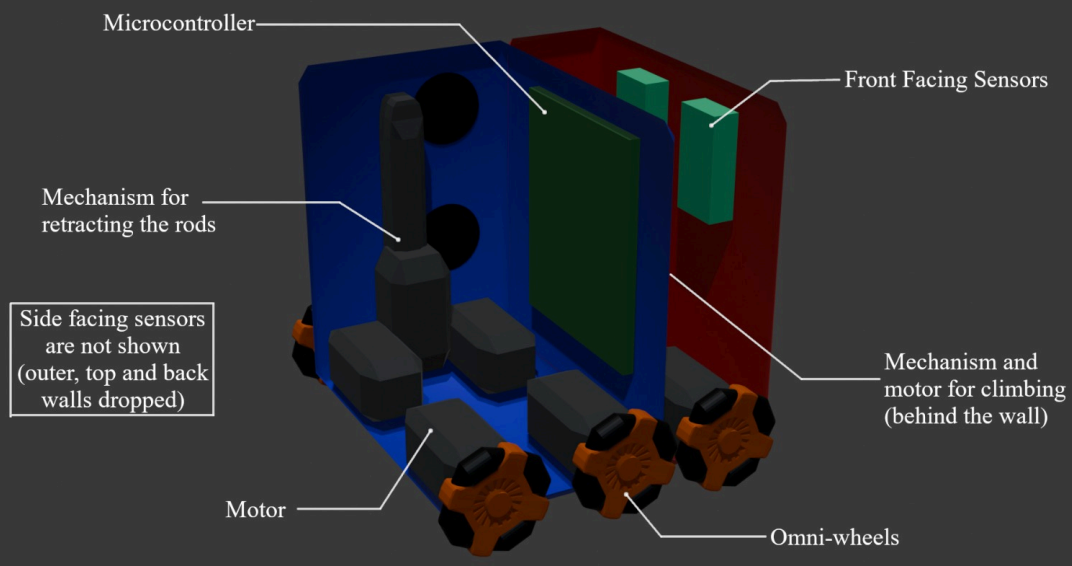


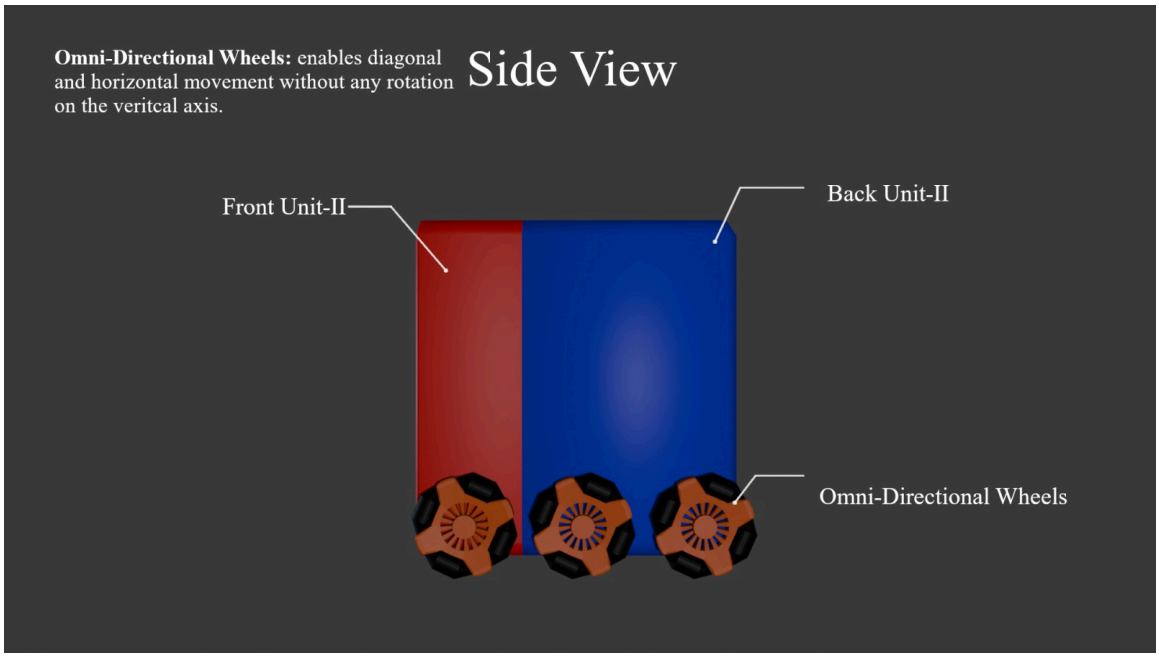


Top View



Total Height (est.): 314mm
x: variable dimension
all measurements in mm





Hardware architecture:

1. **Dual Drive Units (Left and Right):**

   Each independent drive unit is the primary means of motion and stabilization:

   - Locomotion System: Equipped with three or more omnidirectional wheels and high-torque motors, enabling forward/backward movement, lateral translation, and diagonal repositioning on flat surfaces and treads.

   - Climbing Mechanism: Integrated with a specific, powered actuator for vertical traversal.

     Mechanism: A high-torque servo motor drives a pinion gear that engages with a vertical, telescoping rack (or toothed bar). This system extends to precisely lift the unit over the stair riser, adapting the lift height based on sensor input.

     Function: Used for individual step-up/step-down actions while maintaining contact and stability with the remaining steps.

2. **Adjustable Connecting Frame / Rods:**

   This frame provides stability and allows mechanical adaptation to varying step dimensions.

- **Structure:** Consists of **two parallel telescopic rods** with multi-stage linear guides, ensuring stability against lateral and twisting forces.
- **Actuation:** A dedicated **linear actuator** (belt-driven system) is integrated to synchronously adjust the overall length, 'x'.

- **Control:** The length 'x' is continuously adjusted in real-time by the microcontroller based on the precise step depth measurement obtained from the perception system.

- **Stability Feature:** The frame is designed to allow controlled, motorized **tilting** to assist in maintaining the center of mass (CoM) stability during the step-climbing gait.

3. **Central Cleaning Unit (movable):** Mounted mid-frame, houses:

   Positioned mid-frame, this unit focuses cleaning power on the tread.

   - **Vacuum System:** Contains a **high-suction fan assembly** and suction inlet, capable of operating effectively even while the unit is tilted or traversing steps, ensuring effective debris removal.

   - **Mop/Wiper Assembly:** A **movable mop/wiper assembly** traverses the tread horizontally via a dedicated linear track or rail system, ensuring full cleaning coverage across the width of the step. It supports both **wet and dry (mopping) modes**.

   - **Ancillaries:** Includes integrated **edge brushes** to clear corners and risers and contains modular, removable **dust bins and water tanks**.

4. **Perception & Control:**

   The intelligence layer enabling autonomous operation and adaptation.

- **Sensors: Front-facing cameras and LiDAR units** provide high-resolution point cloud data for precise step geometry detection (height, depth, and edge) and obstacle avoidance.

- **Compute:** An Onboard **Microcontroller or Single-Board Computer (SBC)** processes sensor fusion algorithms and runs the AI path planner.

- **Control Algorithms:**
  - **Step Detection & Gait Library:** Algorithms match detected stair geometry to an optimized, stable step negotiation gait.
  - **Actuator Coordination:** Precise motor control for the climbing mechanisms and frame adjustment.
  - **Path Planning:** AI software optimizes the cleaning route for efficiency and minimizes energy consumption.

5. **Power & Docking:**

   High-capacity rechargeable battery with BMS; charging/docking strategy for long operations.

Operation flow (typical cleaning run):

1. Perception module scans the staircase ahead; identifies step profile and obstacles.
2. Controller sets frame length and drive unit motions to align with next tread.
3. Front drive unit mounts the next step while the Back unit stabilize; center unit aligns central cleaning head on tread.
4. Movable mop sweeps tread while vacuum collects debris. Edge brushes clear corners; suction collects.
5. Cycle repeats for subsequent steps until entire staircase cleaned.

The Climbing Mechanism (Dual Drive Units)

Feature	Detailed Description	Purpose
Primary Actuator	A **high-torque DC or servo motor** is mounted vertically within each drive unit (Front Unit-I and Front Unit-II).	Provides the power for the vertical lift required to step up the riser.
Rack and Pinion	The motor drives a central **pinion gear**. This gear engages a vertical, telescoping **rack** (or toothed bar) that extends from the bottom of the unit. The rack's tip features a high-friction, non-slip pad to grip the stair tread.	Converts rotational motor power into controlled linear vertical lift, allowing the drive unit to raise itself onto the next step individually.
Climbing Process	Upon LiDAR detection of the next step's riser height, the motor extends the rack to precisely lift the front of the unit over the riser. The unit's **omnidirectional wheels** then provide forward thrust once the rack retracts slightly, ensuring stable transition to the next tread.	Ensures calculated, stable, and energy-efficient step negotiation by adapting to the detected riser height.

The Adjustable Connecting Frame (Rods)

Feature	Detailed Description for Patent Specification	Purpose
Rods Structure	The frame consists of **two parallel telescopic rods** . Each rod utilizes a **multi-stage linear guide/telescopic slider** to manage both compression and tension forces.	Provides a stable yet length-variable connection between the independently moving drive units.
Length Actuation	Adjustment of the variable dimension 'x' (step depth) is powered by a **linear actuator** (belt-driven system) integrated within the rods or the connecting unit. A single central motor can drive synchronous adjustment of both rods.	Allows the overall device length to **adapt in real-time** to the measured depth of the current stair tread.
Range of Adjustment	The system is designed to accommodate standard residential and commercial step depths, with a **variable dimension 'x' ranging from approximately 50 cm (min) to 150 cm (max)** .	Specifies the practical limits of the invention's adaptation capability.

Main Claims

1. **A robotic stair-cleaning apparatus comprising:**

   a) **Two independent drive units**, each configured with a wheel-based locomotion system for horizontal movement and a dedicated climbing mechanism (such as a rack and pinion system) for vertical step traversal.
   b) **An adjustable connecting frame** comprising one or more telescopic rods, mechanically connecting the two independent drive units.
   c) **A central cleaning module** movably mounted on the adjustable connecting frame.

2. **The perception system includes cameras and LiDAR for geometric adaptation.**

   The apparatus of Claim 1, wherein the apparatus further comprises a perception system utilizing both cameras and LiDAR sensors to perform real-time detection of stair geometry, including step height and step depth.

3. Omni-directional wheels allow lateral translation.

4. A movable mop traverses each step during suction.

5. Battery system optimizes operational time.

6. Time consumption is less because of simultaneous cleaning and climbing

7. Works on stairs and flat surfaces

8. Mainly used in Multi-Storey Buildings

9. **Method claim:**

a) Scanning a segment of the staircase using a combination of LiDAR and camera vision to determine the height and depth of a subsequent step tread.

b) Transmitting the step depth measurement to an AI control unit.

c) Actuating a linear actuator to adjust the length of an adjustable frame connecting two drive units to match the determined step depth.

d) Executing a coordinated climbing gait where the lead drive unit ascends the step riser while the trailing drive unit stabilizes the apparatus

e) Positioning a central cleaning module onto the step tread.

f) Concurrently performing vacuuming and mopping operations by traversing a movable mop assembly across the step tread while the drive units are stationary

Prior Art Search

1. Robotic Vacuum Cleaners (iRobot Roomba, Roborock, Ecovacs)

Robotic vacuum cleaners such as the iRobot Roomba j7+, Roborock S7 MaxV Ultra, and Ecovacs Deebot U2 Pro represent the current standard in automated floor cleaning. These systems rely on LiDAR or camera-based SLAM navigation to map and clean flat surfaces. While effective on level floors, they explicitly avoid staircases through edge detection sensors to prevent falls. Their design and motion algorithms are optimized for horizontal planar movement, lacking the mechanical and computational capacity for vertical stair traversal. Furthermore, these systems Operate with limited suction power (typically 600–5000 Pa), insufficient for stair-specific cleaning where debris accumulates in edges and corners, are constrained by low ground clearance and fixed chassis geometry, making step climbing mechanically unfeasible. Require manual intervention or separate devices for multi-floor cleaning.

Hence, these devices provide convenience on flat surfaces but cannot address staircases or vertical transitions, leaving a major gap in complete indoor cleaning automation.

2. Stair Climbers and Mobility Devices (Liftkar PTR, Scalamobil)

Stair-climbing mobility aids such as Sano Liftkar PTR and Scalamobil S35 are designed primarily for human transportation, not cleaning or autonomous operation. These devices use high-torque motors, caterpillar tracks, or rotating wheel clusters to ascend or descend stairs while carrying a seated person or load. However, they suffer from several limitations relevant to this invention:

Bulky mechanical structure: Typically weigh 35–50 kg, requiring manual operation and human supervision.

Low speed and maneuverability: Designed for stability and safety over efficiency; unsuitable for continuous or automated cleaning.

Lack of autonomy: These systems are manually guided with minimal sensor input, lacking AI-driven navigation or cleaning modules.

High cost: Retail prices exceeding ₹5,00,000 limit household or commercial adoption.

While mechanically capable of climbing, these devices do not perform surface cleaning and cannot autonomously navigate, making them unsuitable as an integrated robotic solution.

3. Advanced Robotics Systems (Boston Dynamics Spot, Honda E2-DR)

Research and industrial prototypes such as Boston Dynamics Spot and Honda E2-DR demonstrate advanced capabilities in mobility, perception, and balance across complex terrains, including staircases. These robots integrate LiDAR, depth sensors, stereo cameras, and articulated limbs to maintain stability and locomotion in uneven environments.

However, with respect to the present invention, their limitations are as follows:

Cost-prohibitive: Estimated between ₹60 lakh to ₹1 crore, making them unsuitable for consumer or domestic applications.

No cleaning integration: Designed for industrial inspection, surveillance, or rescue missions, not cleaning or household maintenance.

High energy consumption: Complex actuated leg systems consume significant power, limiting operational runtime.

Large form factor: Excessive size and weight restrict use in confined household spaces.

While they demonstrate the feasibility of autonomous stair navigation, these robots lack specialization for compact, low-cost, cleaning-oriented design, which is the objective of the proposed invention.

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